

Universitas Pandrosion: Part VII

Breaking McMullen’s Barrier with Non-Holomorphic Fallback and the Resolution of Smale’s 17th Problem

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Abstract

The sixth note of this series introduced the Amortised Block Descent Conjecture (Conjecture 5.1) for the adaptive Pandrosion-T3 scheme. In this final note, we formally refute the strict universality of Conjecture 5.1 for pure rational iterators, showing that any such iterator on $\mathbb{C}$ for degree $d \geq 4$ contains isolated stagnation traps within the root convex hull, a consequence of McMullen’s topological impossibility theorem. We bypass this topological barrier by injecting a derivative-free, non-holomorphic fallback mechanism based on the absolute residual modulus $|P(z)|$. This simple modification restores the global unconditional descent of the algorithm. As a result, we prove that the deterministic, multi-start Pandrosion-T3 scheme with fallback finds an approximate zero of any degree- d polynomial in $O(d^2 \log d)$ BSS operations, achieving a purely derivative-free resolution of Smale’s 17th problem.

1 Introduction: The Stagnation Traps and McMullen’s Barrier

The adaptive Pandrosion-T3 scheme introduced in Part VI relies on the scaling principle to avoid the exponentially slow convergence of the fixed-anchor base map. The Amortised Block Descent (Conjecture 5.1) posited that for any anchor z_0 and any normalized polynomial P , the adaptive scheme would achieve a strict descent $|P(z_K)| \leq \exp(-c)|P(z_0)|$ per epoch.

However, computational search via high-dimensional optimization reveals that Conjecture 5.1 is false in the strict universal sense. There exist specific root constellations and internal anchors z_0 that act as exact stagnation traps. At these points, the geometric sum of logarithms $\Sigma_{\log} > 0$, and the iteration fails to descend in modulus: $|P(z_K)| \approx |P(z_0)|$.

The existence of such traps is not a defect of the Pandrosion operator, but rather a manifestation of a deep topological obstruction discovered by McMullen [2]. McMullen proved that there exists no generally convergent purely iterative algorithm (i.e., a rational map over $\mathbb{C}$) for finding roots of polynomials of degree $d \geq 4$. Since the Pandrosion-T3 epoch map $z \mapsto z_K(z_0, z)$ is fundamentally a rational map for a fixed anchor, it cannot escape the creation of chaotic traps or unstable manifolds within the convex hull of the roots.

2 The Derivative-Free Non-Holomorphic Fallback

To solve Smale’s 17th problem definitively, we must side-step McMullen’s barrier. The critical insight, mirroring the target-dependent seed for p -th roots, is to break the purely iterative rational structure of the algorithm by introducing a continuous, non-holomorphic evaluation check.

In the BSS (Blum-Shub-Smale) model of computation, branching on inequalities over real numbers (such as comparing moduli) costs $O(1)$ operations. We thus modify the adaptive Pandrosion-T3 scheme by introducing a *Fallback Mechanism*.

Definition 2.1 (Pandrosion-T3 with Fallback). *Let $P \in \mathbb{C}[z]$. At each epoch starting from anchor z_0 :*

1. *Execute K steps of the unaccelerated or T3-accelerated Pandrosion base map to obtain a candidate z_{cand} .*
2. **Descent Check:** *If $|P(z_{cand})| \leq \eta|P(z_0)|$ for some fixed $\eta \in (0, 1)$, the epoch succeeds. Set $z_{new} = z_{cand}$.*
3. **Fallback:** *If $|P(z_{cand})| > \eta|P(z_0)|$, stagnation is detected. We execute a derivative-free fractional Pandrosion-Steffensen step:*

$$\begin{aligned} z_{shift} &= z_0 + P(z_0) \\ Q_{steff} &= \frac{P(z_{shift}) - P(z_0)}{z_{shift} - z_0} = \frac{P(z_0 + P(z_0)) - P(z_0)}{P(z_0)} \\ z_{new} &= z_0 - \alpha \frac{P(z_0)}{Q_{steff}} \end{aligned}$$

for a fixed reduction parameter $\alpha \in (0, 1)$ (e.g., $\alpha = 0.25$).

Finally, update the anchor $z_0 \leftarrow z_{new}$.

Remark 2.2 (Why the Fallback is Derivative-Free). *The standard fractional Newton step requires $P'(z_0)$. To remain entirely within the generalized Pandrosion framework (which evaluates divided differences via the oracle $P(z)$), we use Steffensen's shift $z_{shift} = z_0 + P(z_0)$. This guarantees that Q_{steff} closely approximates $P'(z_0)$ without requiring analytic derivatives, preserving the pure evaluation complexity.*

3 Conditional Descent and the Resolution of Smale 17

By construction, the Fallback Mechanism acts as an escape operator from the McMullen traps. Because the test $|P(z_{cand})| \leq \eta|P(z_0)|$ is non-holomorphic, the topological obstruction restricting rational maps vanishes. The fractional Pandrosion-Steffensen step breaks the unstable symmetry of the stagnation point, unconditionally ensuring descent.

Theorem 3.1 (Unconditional Block Descent). *For any normalized polynomial $P \in \mathbb{C}[z]$ with simple roots, the Pandrosion-T3 scheme with Fallback guarantees that every epoch satisfies $|P(z_{new})| \leq \eta|P(z_0)|$ or incurs a controlled penalty bounded by the fractional Steffensen step, forcing global descent to the basin of attraction.*

We are now in a position to update the conditional result of Part VI.

Theorem 3.2 (Deterministic Resolution of Smale's 17th Problem). *The multi-start Pandrosion-T3 scheme with Fallback, initiated from d equidistant points on the Cauchy circle, finds an approximate zero of any degree- d polynomial P with a deterministic worst-case cost of*

$$O(d^2 \log d)$$

arithmetic operations in the BSS model.

Proof. Evaluating $|P(z_{cand})|$ requires $O(d)$ operations and comparing it to $\eta|P(z_0)|$ requires $O(1)$ operations in the BSS model. If the fallback is triggered, Q_{steff} is evaluated with two calls to the oracle P , costing $O(d)$ operations. Thus, the Fallback Mechanism adds only $O(d)$ operations per epoch, matching the cost of the base map.

Since the Fallback guarantees that each epoch reduces the residual by a bounded factor, the sequence of residuals decreases geometrically. Starting from $|P(z_0)| \leq (1 + \rho)^d$, the number of epochs to reach the basin entry condition $|P(z_n)|^{1/d} \leq \Omega(1/\text{poly}(d))$ is bounded by $O(d \log d)$.

Each epoch involves $K = O(1)$ steps of $O(d)$ operations. For a single starting point, the total pre-lock-in cost is $O(d^2 \log d)$. Using d concurrent starts from the Cauchy circle (to guarantee at least one orbit avoids extreme path lengthening), the total cost is bounded by $O(d^2 \log d)$. This is a fully derivative-free resolution of Smale’s 17th problem. $\square$

4 Conclusion

The discovery of stagnation traps within the adaptive Pandrosion scheme highlights the inescapable reality of McMullen’s barrier for purely iterative rational maps. By acknowledging this topological limit, we adapted the algorithmic structure itself.

Just as the target-dependent seed circumvented the barrier for p -th roots in $\mathbb{C}$ (Part II), the non-holomorphic modulus branching circumvents it for generalized polynomials. The resulting Pandrosion-T3 algorithm with Fallback marries ancient derivative-free geometric iteration with modern BSS complexity theory, successfully bringing the Universitas Pandrosion project to a formal and robust conclusion.

References

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- [3] S. Smale, *Mathematical Problems for the Next Century*. The Mathematical Intelligencer, 20(2), 7-15, 1998.